

AKASHRAJ KALATHINGAL

Senior Software Engineer | Full-Stack • Distributed Systems • Payments Infrastructure • AI Platforms

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with 6+ years of production experience spanning high-value payments infrastructure, AI-powered banking products, and petabyte-scale enterprise data orchestration. Deep expertise in Java/Spring Boot microservices, event-driven distributed systems, and modern frontends in React and Angular. Hands-on architect of LLM RAG systems using MCP tooling, ISO 20022 payment modernization, and multi-cloud migrations across AWS, Azure, GCP, and OpenShift.

Known for taking ambiguous problems from spec to production: I write the design doc, ship the service, own the on-call rotation, and mentor the next engineer. Excited about backend, platform, and infrastructure problems at companies that scale.

CORE COMPETENCIES

Distributed Systems • Microservices Architecture • Event-Driven Systems • API Design • System Design • Scalability • Reliability Engineering • Observability • Cloud-Native Development • Kubernetes Orchestration • CI/CD Automation • Security & Compliance • LLM & RAG Systems • Payments Engineering • Data Engineering • Technical Leadership • Cross-Functional Collaboration • Mentorship & Code Review

TECHNICAL SKILLS

Programming Languages: Java (8 / 11 / 17), TypeScript, JavaScript (ES6+), Python, SQL, Scala, GraphQL, Bash

Backend & Frameworks: Spring Boot, Spring MVC, Spring Batch, Spring Security, REST, gRPC, GraphQL, Kafka, RabbitMQ, OAuth2 / JWT, JPA / Hibernate, Microservices, Event-Driven Architecture, Domain-Driven Design

Frontend: React (Hooks, Router, Context API), Angular (12+), Redux, Material UI, Tailwind, HTML5, CSS3, Webpack, Vite, npm / yarn

Cloud Platforms: AWS (S3, EKS, EC2, Lambda, IAM), Microsoft Azure (AKS, Document Intelligence, DevOps), Google Cloud Platform (GCP)

Containers & Infra: Docker, Kubernetes, Red Hat OpenShift, Helm, Ansible, HashiCorp Vault, Terraform-style IaC, Windows / Linux On-Prem Servers

Databases: PostgreSQL, Oracle, IBM DB2, MongoDB, Neo4j, Apache Cassandra (familiarity)

Data & Analytics: Apache Hadoop, Apache Hive, Apache Spark, Apache Druid, Kafka Streams, ETL pipelines, Kedro

CI/CD & DevOps: Azure DevOps, Jenkins, Drone, GitHub Actions, Bitbucket Pipelines, Maven, Gradle, Git, SonarQube

Observability & Monitoring: Grafana, Kibana, ELK Stack, Splunk, Filebeat, DOMO, PagerDuty-style on-call workflows

AI / ML Engineering: LLM RAG Pipelines, Model Context Protocol (MCP), Vector Embeddings, Kedro, Azure Document Intelligence, scikit-learn, pandas, NumPy, OpenCV

Payments & Financial: IBM Financial Transaction Manager (FTM), IBM Integration Bus (IIB), IBM MQ / Queue Manager, IBM WebSphere, IBM JBoss, IBM Rational Software Architect, IBM iKeyman, SWIFT, ISO 20022 (MX Messaging), Wire Payments, FX Conversion, Fraud Validation

Testing & Quality: JUnit, Mockito, Cypress, Postman, Integration Testing, Load Testing, Regression Testing, Usability Testing, SonarQube static analysis

Security & Compliance: OAuth2, JWT, SSL / TLS certificate management, HashiCorp Vault, Log4Shell remediation (CVE-2021-44228), PCI-aware engineering, SOC2-aware practices

Architecture & Methods: System Design from Scratch, Finite State Machines, Wireframing, SPIKE-driven design, Release Management, Agile / Scrum, SAFe-style cross-team delivery

PROFESSIONAL EXPERIENCE

Selected Contributions

- Core engineer on BMO InvestorLine's **AI-driven financial advisory chatbot** that delivers portfolio analysis, buy/sell recommendations, and personalized advice to retail investors. Designed and implemented Java/Spring Boot microservices that expose internal banking capabilities as **MCP (Model Context Protocol) tools** for LLM RAG pipelines, enabling tool-using agents to securely orchestrate advisory workflows.
- Designed end-to-end **data ingestion pipelines using Kedro and Azure Document Intelligence** to extract, normalize, and embed unstructured financial documents (prospectuses, statements, regulatory filings) into vector stores powering the chatbot's retrieval layer.
- Led **secrets management migration from Red Hat OpenShift to HashiCorp Vault** across 10+ production services, replacing in-cluster secret stores with Vault-issued dynamic credentials and short-lived tokens — strengthening compliance posture and reducing blast radius of credential exposure.
- Drove **on-prem to AWS migration** of critical production services, re-platforming workloads onto containerized AWS infrastructure with improved scalability, multi-AZ resilience, and IaC-driven deployments.
- Automated CI/CD pipelines with **Azure DevOps and Ansible**, standardizing release management across multiple services and reducing manual deployment toil for the team.
- Built **modular, accessible Angular components** for AdviceDirect onboarding flows including multi-account enrollment, KYC, and risk-profile capture — improving conversion and consistency across self-directed and assisted-investing journeys.
- Delivered RESTful APIs integrating with internal banking systems for account data, market data, transactions, and customer profiles, maintaining strict SLAs on availability and latency.
- Operated services on **Red Hat OpenShift** with horizontal autoscaling and zero-downtime rollouts across enterprise Kubernetes clusters; partnered with SRE on capacity planning and incident response.
- Collaborated with product owners, UX designers, QA, and platform teams in an Agile environment to ship customer-facing features aligned with BMO's digital modernization roadmap.

Tech Stack: Java 17, Spring Boot, Angular, Kedro, Azure Document Intelligence, MCP, OpenShift, AWS, HashiCorp Vault, Azure DevOps, Ansible, Docker

Scotiabank

Jan 2024 – Feb 2025

Software Engineer — High-Value Payments & Core Banking Platform Toronto, ON, Canada**Selected Contributions**

- Acted as both **IBM Financial Transaction Manager (FTM) developer** and **Java Full-Stack engineer** on Scotiabank's next-generation high-value payments platform, building and enhancing systems that handle outgoing wire transfers, SWIFT messaging, and FX transactions for Canadian and cross-border banking customers.
- Led the bank's **ISO 20022 migration program** for outgoing MX-format wire payments. Designed planning matrices, performed system impact analyses, and shipped automated flows for fraud validation, FX conversion, and charge posting — eliminating manual operator intervention and aligning Scotiabank with global SWIFT mandates.
- Owned **core banking security remediation**: renewed and managed SSL/TLS certificates across multiple payment applications and platforms, established lifecycle tracking, and prevented service disruption from expirations on systems handling regulated transaction flows.
- Architected fraud-validation integration with the bank's central payment hub, defining contracts between FTM, fraud engines, and clearing systems to ensure low-latency, deterministic decisions on time-sensitive wire payments.
- Evaluated interrelationships across the high-value payment ecosystem — payment hubs, fraud engines, SWIFT gateways, FX engines, charge engines — and provided expert recommendations on whether to extend existing systems or stand up new ones, reducing technical debt while accelerating delivery.
- Provided **front-line production support** on a rotating on-call schedule, triaging incidents on systems with strict business-hour SLAs and zero tolerance for transaction loss; authored post-mortems and drove remediation backlog.

- Authored comprehensive technical and functional design documents, runbooks, and knowledge bases — directly reducing ramp time for new team members and serving as canonical references during audits.

Tech Stack: Java, Spring Boot, IBM FTM, IBM IIB, IBM MQ, IBM WebSphere, Oracle, SWIFT MT/MX, ISO 20022, SSL/TLS

Target Corporation

Dec 2019 – Apr 2023

Senior Software Engineer (promoted from Software Engineer) — Enterprise Orchestration Platform Bangalore, India

Selected Contributions

- Designed and scaled core microservices for Target's **enterprise data orchestration and scheduling platform**, supporting petabyte-scale data movement and workflow orchestration across heterogeneous systems including Hadoop, Hive, on-prem databases, and cloud object stores.
- Architected and shipped enhancements to the platform's **Patroller subsystem** — a self-healing job-recovery engine that detected stuck and failed workflows, retried with backoff, and escalated. Materially improved job success rates and reduced operator pages across peak retail seasons (Black Friday / Cyber Monday).
- Built the platform's **reporting and analytics layer** on Apache Druid + Grafana, giving leadership real-time visibility into throughput, error rates, and capacity utilization — directly informing platform investment and capacity-planning decisions.
- Drove **platform-wide Log4Shell (CVE-2021-44228) remediation** across the orchestration estate, coordinating dependency upgrades, regression testing, and emergency releases under tight executive timelines.
- Modernized the platform's admin and operator UI in **React (Hooks, Context API, Material UI)**, delivering reusable component libraries for job scheduling, monitoring, and configuration — adopted by hundreds of internal engineers.
- Authored production APIs across multiple codebases and led **Druid adapter parameter tuning** to optimize ingestion throughput and query latency.
- Stood up **CI/CD pipelines using Drone, Jenkins, Docker, and Kubernetes** to standardize deployments and improve release velocity for the team.
- Mentored and **onboarded junior engineers**, running code review walkthroughs, pairing on production incidents, and contributing to team-wide engineering standards.
- Led peak readiness planning and chaos / load testing strategy that ensured platform stability during Target's highest-traffic retail events of the year.
- Resolved multiple critical production incidents including complex Hive workflow failures and Kibana logging pipeline outages from scheduler nodes.
- Recognized with the **Target Data Science Hackathon win**, multiple Pyramid Performance Awards, and the Quarterly Best Performer Award.

Tech Stack: Java, Spring Boot, React, Apache Druid, Apache Hive, Apache Spark, Kafka, MongoDB, Oracle, Kubernetes, Docker, Drone, Jenkins, Grafana, Kibana, ELK

CGI Inc.

Aug 2018 – Dec 2019

Software Engineer Bangalore, India

Selected Contributions

- Built an **enterprise ERP web application** and an event-hosting platform end-to-end, owning backend services (Java/Spring), relational data modeling, REST APIs, and frontend delivery.
- Investigated and resolved high-severity production defects under strict client SLAs, producing reproducible repro steps and root-cause analyses.
- Authored production-grade technical documentation and reproducible runbooks adopted as onboarding material for new engineers.

Tech Stack: Java, Spring, JavaScript, HTML/CSS, SQL

SELECTED ENGINEERING PROJECTS

AI-Powered Financial Advisory Chatbot (BMO InvestorLine)

Customer-facing LLM agent for personalized investment advice across BMO retail accounts.

- Designed Java/Spring Boot microservices exposing banking APIs as **MCP tools** consumed by an LLM agent, with structured input/output schemas, auth scopes, and audit logging.
- Built ingestion pipelines with Kedro and Azure Document Intelligence to extract entities and embeddings from prospectuses, statements, and regulatory filings for RAG retrieval.
- Engineered guardrails, prompt routing, and tool-use orchestration patterns enabling deterministic, auditable advisory responses on regulated banking data.

ISO 20022 Wire Payment Automation (Scotiabank)

Modernization of outgoing high-value wire payments to align with global SWIFT 2025 standards.

- Implemented automated processing of MX-format wire messages with fraud validation, FX conversion, and charge posting — replacing manual operator workflows.
- Designed planning matrices and impact analyses for ISO 20022 enablement across the high-value payments stack, partnering with risk, fraud, and SWIFT operations teams.
- Hardened SSL/TLS certificate lifecycle management on the payment platform, preventing service-affecting expirations.

Patroller — Self-Healing Job-Recovery Engine (Target)

Reliability subsystem for Target's enterprise orchestration platform.

- Built detection, retry, and escalation logic for stuck and failed workflows, with exponential backoff and dead-letter queues.
- Reduced operator paging load and improved end-to-end job success rates during Target's highest-throughput retail events.

Druid + Grafana Reporting Layer (Target)

Real-time analytics for platform health and capacity planning.

- Modeled Druid datasources for sub-second slice-and-dice across throughput, latency, and error metrics.
- Tuned Druid ingestion adapters and Grafana dashboards adopted by leadership for portfolio-level decisions.

Log4Shell (CVE-2021-44228) Platform-Wide Remediation (Target)

- Coordinated identification, upgrade, regression-testing, and emergency release of all affected services on a compressed timeline.
- Established follow-on dependency-hygiene practices including SCA scans and SBOM tracking.

EDUCATION

Conestoga College — Post Graduate Certificate, Big Data Solutions & Architecture

May 2023 – Dec 2023

Waterloo, ON, Canada

- Graduated with **High Distinction**; named to the Dean's Honour List as the top performer in the cohort.
- Coursework: Distributed Computing, Apache Spark, Hadoop, Cloud Architecture, Data Engineering, Machine Learning at Scale.

UT Austin, McCombs School of Business — Dual Certificate PG Program, AI & Machine Learning

May 2020 – May 2021

Coursework spanning supervised / unsupervised learning, deep learning, NLP, and applied ML systems.

National Institute of Technology (NIT) Calicut — Bachelor of Technology, Computer Science & Engineering

2018

Calicut, Kerala, India • One of India's premier engineering institutes (top-tier NIT, government-of-India accredited).

AWARDS & RECOGNITION

- Winner — **Target Corporation Data Science Hackathon**
- Pyramid Performance Award (awarded multiple times) — Target Corporation & Scotiabank
- Quarterly Best Performer Award — Target Corporation
- Dean's Honour List (Top Performer) — Conestoga College